

Locomotive: RENFE 333.4 | **Track:** Level (CTN test route) | **Speed:** 100 → 0 km/h (emergency brake) | **Date:** May 2026

1. Introduction

Pneumatic brake tests are conducted using emergency braking with different types of brake shoes, including cast iron, composite material, disc brakes, and custom curves. The locomotive's speed is increased to 100 km/h and braking is applied, measuring distances for each type used in the ORTSBrakeShoeType parameter.

2. Materials

All tests were performed on level track using a single locomotive, specifically one belonging to the 333.4 series, previously calibrated and with Open Rails anti-lock system enabled:

- ORTSWheelBrakeSlideProtection (1)
- ORTSEmergencyBrakingDisablesWSP (0)

The route used belongs to the CTN test route (<https://www.coalstonewcastle.com.au/physics/route/>). The specific path is one included with the route, namely "Outer Loop - NE Corner Station / Outer Loop - SW Corner Station" within the "Long outer circuit - Level" layout. Simulation software: Open Rails New Year MG 171.3 stable.

3. Description of the Empirical Method

To carry out the experiment, the ORTSBrakeShoeType parameter is used, which allows defining "a number of different brake shoes and braking curves" according to the Open Rails manual. This parameter has five possible states:

State	Description
Cast_Iron_P6	Older cast iron brake shoes
Cast_Iron_P10	Phosphoric cast iron shoes, more modern than the previous ones
Hi_Friction_Composite	High-friction composite shoes
Disc_Pads	Brake disc pads
User_Defined	Custom curve defined via ORTSBrakeShoeFriction

Table 1: Brake shoe types available in Open Rails via the ORTSBrakeShoeType parameter. Source: Open Rails 1.6.1 Manual (2026).

The custom curve used for the User_Defined state was:

```
0.0 0.49 8.0 0.436 16.1 0.4 24.1 0.371 32.2 0.35 40.2 0.336 48.3 0.325 56.3 0.318
64.4 0.309 72.2 0.304 80.5 0.298 88.5 0.295 96.6 0.289 104.6 0.288
```

Procedure:

Once the brake shoe type is selected, the locomotive is loaded in the simulator and its speed is increased to 100 km/h, at which point the emergency brake is applied (full reverser) and the throttle is closed. During the tests, the throttle did not exceed 50% of its application to interfere as little as possible with braking performance.

The distance between brake application and full locomotive stop is measured. The reading is recorded on the data sheet. In cases where wheel lock occurs, the distance during which the wheels are locked is measured.

4. Experimental Results

Below is a table with the results obtained:

Brake Shoe Type	Braking Distance (m)	Observations
Cast_Iron_P6	741	No wheel lock
Cast_Iron_P10	652	Wheel lock in final metres (< 20 m)
Hi_Friction_Composite	525	Wheel lock in final 44 metres. Used in the locomotive by default
Disc_Pads	487	Wheel lock in final 13 metres
User_Defined	517	Wheel lock in final 87 metres

Table 2: Braking distances from 100 km/h to full stop using emergency braking on level track. Wheel lock distances measured visually from the HUD. Locomotive: RENFE 333.4. Dry conditions, WSP enabled.

5. Conclusions

- As can be observed in the table, the distances obtained are consistent with the function of the selected state for the ORTSBrakeShoeType parameter, with braking distance decreasing alongside the evolution of materials used, as expected.

- The curve chosen for the User_Defined state is susceptible to adjustment with data available from manufacturers.

6. References

- jtr1962 (Elvas Tower member). Guideline For Setting Open Rails Braking Parameters (2019)
- Open Rails Team. Open Rails 1.6.1 Manual, Release 1.6.1.0 (2026)

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